An algebraic explanation for the degree threshold 9/8 in Fox–Pach–Suk's bound towards Albertson's conjecture

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Abstract

In their proof of Lemma 2.3 towards Albertson's conjecture (arXiv:2510.05893, SoCG 2025), Fox, Pach, and Suk introduce a degree threshold $d := \delta k$ and fix $\delta = 9/8$ without justification. We show that this choice is sharp within their Vizing–Gupta plus semi-random framework. Concretely, defining $F(\delta)$ to be the supremum of the three-case Claim 3.7 objective at a given δ, we prove that

$$F(\delta) \;\geq\; \tfrac{9}{16} \qquad \text{for every } \delta \in (1,\, 5/4),$$

with equality if and only if $\delta = 9/8$. The proof of the strict inequality $F(\delta) > 9/16$ for $\delta \in (1,\, 9/8)$ is analytic and short: at the witness point $\eta = 4/7$ in Case 2b,

$$f_{2b}\big(\tfrac{4}{7},\, \delta\big) - \tfrac{9}{16} \;=\; \frac{12\,(\delta - 9/8)^2}{7\,(4\delta - 1)} \;>\; 0 \qquad \text{for } \delta \neq 9/8.$$

Two structural reasons underlie this sharpness. First, the Case 2b objective $f_{2b}(\eta, \delta)$ has nonpositive derivative in the reparametrising variable η at $\eta = 1/2$ if and only if $\delta \geq -3 + \sqrt{17} \approx 1.12311$, with strict negativity for $\delta > -3 + \sqrt{17}$; the FPS choice $\delta = 9/8 = 1.125$ sits just above this threshold. For $\delta < -3 + \sqrt{17}$ the derivative is positive at $\eta = 1/2$ and the maximum of f_{2b} moves into the interior of its η-interval. Second, the Case 1 / Case 2a equalisation line $\delta/2 = f_{2a}^{\max}(\delta)$ leads to the irreducible cubic $\delta^3 + 3\delta^2 - \delta - 4 = 0$, whose root in $(1, 5/4)$ sits at $\delta_1 \approx 1.114907 < -3 + \sqrt{17}$ – so the "naive" cubic optimum falls precisely in the regime where the Case 2b monotonicity assumption used in deriving it is violated.

Any improvement to the Lemma 2.3 constant $c_{\mathrm{FPS}} = 9/16$ must therefore come from outside this FPS Claim 3.7 optimisation as currently formulated – for example, by replacing the Vizing–Gupta bound $\chi'(H) \leq \Delta(H) + \mu(H)$ by Goldberg–Seymour or Kahn-type estimates, or by modifying the semi-random construction itself.

1 Introduction

Fox, Pach, and Suk [3] prove Albertson's conjecture [1], $\mathrm{cr}(G) \geq \mathrm{cr}(K_k)$ for every k-chromatic graph G, in the range $|V(G)| \leq 1.4(k-1)$ (asymptotically $|V(G)| \leq (1.64 - o(1))\,k$). The crucial technical step is their Lemma 2.3, which bounds the chromatic index of an auxiliary multigraph H by

$$\chi'(H) \;\leq\; (c_{\mathrm{FPS}} + o(1))\,k, \qquad c_{\mathrm{FPS}} \;=\; 9/16. \tag{1}$$

The constant 9/16 comes from an explicit three-case optimisation ([3, Claim 3.7]) parameterised by a real degree threshold $\delta := d/k$. Throughout [3], δ is fixed equal to 9/8 from the start; the paper does not justify this choice.

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A natural question is whether $\delta = 9/8$ is forced – i.e. whether the optimum of the three-case bound, treated as a function of δ, is attained at $\delta = 9/8$ – or whether a smaller constant $c < 9/16$ might be reached at some other value of δ. The purpose of this note is to answer that question: $\delta = 9/8$ is essentially forced.

Theorem 1. *Let $F : (1, 5/4) \to \mathbb{R}$ denote the supremum of the Case 1, Case 2a, and Case 2b objectives of [3, Claim 3.7] as a function of the degree threshold δ (see Section 2). Then*

$$F(\delta) \;\geq\; \tfrac{9}{16} \qquad \textit{for every } \delta \in (1, 5/4),$$

with equality if and only if $\delta = 9/8$.

The interval $(1, 5/4)$ is the natural FPS-admissible range: $\delta > 1$ is needed for the Case 2a stationary point to exist, and $\delta < 5/4$ is the upper bound coming from Claim 3.6 of [3] (see Section 2). The Case 2b interval $\eta \in [1/2, \eta_b(\delta))$ is non-empty throughout the FPS-admissible range, becoming empty only at $\delta = 4/3 > 5/4$.

Theorem 1 shows that any improvement to the constant $9/16$ in [3, Lemma 2.3] obtained by retuning δ alone is impossible: such an improvement must come from outside this Claim 3.7 optimisation as currently formulated. Section 7 discusses three concrete routes (replace Vizing–Gupta, sharpen the multiplicity bound, modify the semi-random construction).

The genesis of this note was an erroneous earlier draft (preserved as `tighter_fps_RETRACTED.tex` for historical record) that proposed lowering δ from $9/8$ to a value $\delta_1 \approx 1.114907$ in order to improve $9/16$. The proposed improvement relied on assuming that the Case 2b maximum equals $\delta/2$ at every δ, which is true at $\delta = 9/8$ (FPS prove it) but *not* for general δ. The cubic $\delta^3 + 3\delta^2 - \delta - 4 = 0$ remains an interesting algebraic feature of the Case 1 / Case 2a equalisation (Remark 8), but its root δ_1 is not optimal.

2 Setup

We follow the notation of [3] throughout. Let G be a k-chromatic graph, fix an index i in the Gallai decomposition, and drop subscripts; let $d := \delta k$ be a real degree threshold with $\delta > 1$. Set $\alpha = \ell_w/k$, $\beta = \ell/k$, $\gamma = |U_w|/k$ as in [3, §3]. Reparametrise $\eta := 1 - (1-\beta)/(2-\beta) \in [1/2, 1]$. The Claim 3.7 optimisation maximises

$$f(\alpha, \beta, \gamma; \delta) \;=\; \gamma - \alpha\,\frac{\delta - 1}{\delta - \gamma} \tag{2}$$

over the feasible set

$$0 \leq \alpha \leq \beta \leq 1, \qquad 0 \leq \gamma \leq \alpha + (\delta - \alpha)\,\frac{1-\beta}{2-\beta}. \tag{3}$$

The supremum is denoted $F(\delta)$ and splits into three cases:

Case 1. $\alpha = 0$.

Case 2a. The upper constraint on γ is tight, with the stationary point of f in α interior to $[0, \beta]$.

Case 2b. The upper constraint on γ is tight, with the stationary point falling outside $[0, \beta]$, forcing $\alpha = \beta = 2 - 1/\eta$.

At $\delta = 9/8$, Fox–Pach–Suk's calculation [3, p. 10] gives $F(9/8) = 9/16$, attained by Case 1 at $\beta = 0$ and by Case 2b at $\eta = 1/2$.

Admissible range of δ. We distinguish two ranges. The *FPS-admissible* range is $\delta \in (1, 5/4)$: the lower bound $\delta > 1$ is needed for the Case 2a stationary point to exist, and the upper bound $\delta < 5/4$ comes from [3, Claim 3.6], whose proof requires $4d < 5k$ (i.e. $\delta < 5/4$). The *optimisation-defined* range $\delta \in (1, 4/3)$ is the larger interval on which the Case 2 case-split is non-degenerate; the upper bound $\delta < 4/3$ is the threshold at which the Case 2b interval $[1/2, \eta_b(\delta))$ collapses to a single point (Section 4). On the FPS-admissible range $(1, 5/4)$, Case 2b is always non-empty.

3 The three case bounds with δ free

3.1 Case 1.

With $\alpha = 0$, the active constraint reduces to $\gamma \leq \delta(1-\beta)/(2-\beta)$, which is decreasing in $\beta \in [0, 1]$. The maximum is at $\beta = 0$, giving

$$f_1(\delta) \;=\; \delta/2.$$

3.2 Case 2 stationary point and 2a/2b boundary.

Substituting $\gamma = \eta\alpha + (1-\eta)\delta$ (the active Case 2 constraint) into (2) and differentiating in α shows that the unique stationary point in α is

$$\alpha^\star(\eta, \delta) \;=\; \delta - \frac{\sqrt{\delta(\delta-1)}}{\eta}. \tag{4}$$

The Case 2a regime is $\alpha^\star(\eta, \delta) \leq \beta(\eta) = 2 - 1/\eta$, equivalently $\eta \geq \eta_b(\delta)$ where

$$\eta_b(\delta) \;:=\; \frac{\sqrt{\delta(\delta-1)} - 1}{\delta - 2}. \tag{5}$$

The Case 2b regime is $\eta \in [1/2, \eta_b(\delta))$, in which the optimum is attained at the boundary $\alpha = \beta = 2 - 1/\eta$. At $\delta = 9/8$ this recovers FPS's $\eta_b = 5/7$ and $\alpha^\star = (9 - 3/\eta)/8$ [3, p. 10].

3.3 Case 2a.

In Case 2a, substituting $\alpha = \alpha^\star$ into f gives a function of η alone. A direct calculation shows it is strictly decreasing in η on $[\eta_b(\delta), 1]$, so its maximum is at $\eta = \eta_b(\delta)$. The resulting value is

$$f_{2a}^{\max}(\delta) \;=\; \delta - 2\sqrt{\delta(\delta-1)} + \frac{(\delta-1)(\delta-2)}{\sqrt{\delta(\delta-1)} - 1}. \tag{6}$$

At $\delta = 9/8$ this evaluates to $11/20 = 0.55$, matching [3, p. 10].

4 Case 2b: monotonicity transition at $\delta = -3 + \sqrt{17}$

In Case 2b, $\alpha = \beta = 2 - 1/\eta$, $\gamma = 2\eta - 1 + (1-\eta)\delta$, and $\delta - \gamma = \eta(\delta - 2) + 1$. Substituting into (2),

$$f_{2b}(\eta, \delta) \;=\; \big[2\eta - 1 + (1-\eta)\delta\big] \;-\; \frac{(2\eta-1)(\delta-1)}{\eta\,[\eta(\delta-2)+1]}. \tag{7}$$

At $\eta = 1/2$, $f_{2b}(1/2, \delta) = \delta/2$; at $\eta \uparrow \eta_b(\delta)$, f_{2b} matches $f_{2a}^{\max}(\delta)$ by continuity of the optimisation across the Case 2a/2b boundary.

The decisive observation. The behaviour of f_{2b} near $\eta = 1/2$ as δ varies is controlled by a single elementary identity.

Lemma 2. *For every $\delta > 1$,*

$$\left.\frac{\partial f_{2b}}{\partial \eta}\right|_{\eta=1/2} = -\frac{\delta^2 + 6\delta - 8}{\delta}. \tag{8}$$

Proof. Differentiating (7) in η and evaluating at $\eta = 1/2$ is a direct computation. Writing $A(\eta) := 2\eta - 1 + (1-\eta)\delta$ and $B(\eta) := \eta[\eta(\delta-2)+1]$, we have $A'(\eta) = 2-\delta$, $A(1/2) = \delta/2$, $B(\eta) = (\delta-2)\eta^2 + \eta$, $B'(\eta) = 2(\delta-2)\eta + 1$, $B(1/2) = (\delta-2)/4 + 1/2 = \delta/4$, $B'(1/2) = (\delta-2)+1 = \delta - 1$, and at $\eta = 1/2$ the numerator $(2\eta-1)(\delta-1) = 0$. Hence the derivative of the rational term equals $-(\delta-1)\cdot 2/B(1/2) = -(\delta-1)\cdot 2/(\delta/4) = -8(\delta-1)/\delta$. Adding $A'(1/2) = 2-\delta$ gives

$$\left.\frac{\partial f_{2b}}{\partial \eta}\right|_{\eta=1/2} = (2-\delta) - \frac{8(\delta-1)}{\delta} = \frac{(2-\delta)\delta - 8(\delta-1)}{\delta} = \frac{-\delta^2 - 6\delta + 8}{\delta},$$

which is the negative of $(\delta^2 + 6\delta - 8)/\delta$. ☐

Corollary 3. *$\partial f_{2b}/\partial\eta|_{\eta=1/2}$ is negative iff $\delta^2 + 6\delta - 8 > 0$, iff $\delta > -3 + \sqrt{17}$ (on $\delta > 0$). Hence:*

- *For $\delta > -3 + \sqrt{17}$, f_{2b} is locally decreasing in η at $\eta = 1/2$.*
- *For $\delta = -3 + \sqrt{17}$, $\eta = 1/2$ is a stationary point of f_{2b}.*
- *For $\delta < -3 + \sqrt{17}$, f_{2b} is locally increasing in η at $\eta = 1/2$, and the maximum $f_{2b}^{\max}(\delta)$ is attained strictly in the interior of $[1/2, \eta_b(\delta))$.*

Remark 4. The FPS choice $\delta = 9/8 = 1.125$ exceeds the threshold $-3 + \sqrt{17} \approx 1.12311$ by less than 0.002. This gives a structural explanation for the choice: $\delta = 9/8$ is just inside the regime where the FPS proof of [3, Claim 3.7, Case 2b] (asserting that the Case 2b maximum is attained at $\eta = 1/2$) is valid via the monotonicity-in-η argument they use. For δ just below the threshold $-3 + \sqrt{17}$, the Case 2b maximum slides into the interior of $[1/2, \eta_b(\delta))$ and exceeds $\delta/2$, but *this monotonicity argument is not needed for the main theorem*: the witness identity of Lemma 5 below directly produces a value above 9/16 at an explicit Case 2b point, for every $\delta \in (1, 9/8)$.

5 The witness identity at $\eta = 4/7$

The proof of Theorem 1 for $\delta < 9/8$ rests on the following elementary identity, which exhibits a single explicit witness point $\eta = 4/7$ at which f_{2b} already exceeds 9/16 whenever $\delta \neq 9/8$.

Lemma 5 (Witness identity)**.** *For every $\delta > 1/4$,*

$$f_{2b}\left(\frac{4}{7}, \delta\right) - \frac{9}{16} = \frac{12\,(\delta - 9/8)^2}{7\,(4\delta - 1)}. \tag{9}$$

Proof. A direct computation. At $\eta = 4/7$, $\alpha = \beta = 2 - 7/4 = 1/4$, so $\gamma = (4/7)(1/4) + (3/7)\delta = (1+3\delta)/7$ and $\delta - \gamma = (4\delta - 1)/7$. Hence

$$f_{2b}(\tfrac{4}{7}, \delta) = \frac{1+3\delta}{7} - \frac{(1/4)(\delta-1)\cdot 7}{4\delta - 1} = \frac{1+3\delta}{7} - \frac{7(\delta-1)}{4(4\delta-1)}.$$

Bringing to common denominator $28(4\delta-1)$,

$$f_{2b}\big(\tfrac{4}{7},\delta\big) \;=\; \frac{4(1+3\delta)(4\delta-1)-49(\delta-1)}{28(4\delta-1)} \;=\; \frac{48\delta^2-45\delta+45}{28(4\delta-1)} \;=\; \frac{3(16\delta^2-15\delta+15)}{28(4\delta-1)}.$$

Subtracting $9/16 = 9/16$ from this fraction (over common denominator $112(4\delta-1)$),

$$f_{2b}\big(\tfrac{4}{7},\delta\big)-\tfrac{9}{16} \;=\; \frac{12(16\delta^2-15\delta+15)-9\cdot 7(4\delta-1)}{112(4\delta-1)} \;=\; \frac{192\delta^2-432\delta+243}{112(4\delta-1)}.$$

The numerator factors as $192\delta^2-432\delta+243 = 3(64\delta^2-144\delta+81) = 3(8\delta-9)^2 = 192(\delta-9/8)^2$. Substituting back gives (9). □

Corollary 6. *The witness point* $\eta = 4/7$ *lies in the Case 2b interval* $[1/2,\, \eta_b(\delta))$ *as long as* $\delta < (41+7\sqrt{37})/66 \approx 1.2663$*; in particular for every* δ *in the FPS-admissible range* $(1, 5/4)$*. For every such* δ,

$$f_{2b}^{\max}(\delta) \;\geq\; f_{2b}\big(\tfrac{4}{7},\delta\big) \;\geq\; 9/16,$$

with the second inequality strict whenever $\delta \neq 9/8$.

Proof. The numerator $(\delta-9/8)^2$ in (9) is non-negative, with equality iff $\delta = 9/8$. The denominator $112(4\delta-1)$ is positive for $\delta > 1/4$. Hence the right-hand side is strictly positive whenever $\delta \neq 9/8$ and $\delta > 1/4$, and in particular on the stated δ-interval.

For the validity of the witness, $\eta = 4/7 \in [1/2, \eta_b(\delta))$ iff $\eta_b(\delta) > 4/7$, i.e.

$$\frac{\sqrt{\delta(\delta-1)}-1}{\delta-2} \;>\; \frac{4}{7}.$$

Since $\delta - 2 < 0$ on the relevant range, multiplying both sides by $(\delta-2)$ *flips* the inequality:

$$\sqrt{\delta(\delta-1)}-1 \;<\; \tfrac{4}{7}(\delta-2) \;=\; \tfrac{4\delta-8}{7},$$

equivalently $7\sqrt{\delta(\delta-1)} < 4\delta-1$. For $\delta > 1$ both sides are positive, so squaring preserves the direction:

$$49\,\delta(\delta-1) \;<\; (4\delta-1)^2,$$

which simplifies to $33\delta^2-41\delta-1 < 0$, with positive root $\delta = (41+7\sqrt{37})/66 \approx 1.2663$. Since $5/4 < (41+7\sqrt{37})/66$, the witness is valid throughout the FPS-admissible range. □

Proof of Theorem 1. We split into cases on δ.

The interval $\delta \in [9/8,\, 5/4)$. For every δ in this range we have $f_1(\delta) = \delta/2 \geq 9/16$, with equality only at $\delta = 9/8$. In particular,

$$F(\delta) \;\geq\; f_1(\delta) \;\geq\; 9/16,$$

with equality on this interval iff $\delta = 9/8$.

The interval $\delta \in (1,\, 9/8)$. By Corollary 6, $f_{2b}^{\max}(\delta) > 9/16$, hence

$$F(\delta) \;\geq\; f_{2b}^{\max}(\delta) \;>\; 9/16.$$

At $\delta = 9/8$. By [3, Claim 3.7], $F(9/8) = 9/16$.

Combining the three cases, $F(\delta) \geq 9/16$ on $(1, 5/4)$ with equality iff $\delta = 9/8$. □

Remark 7. At $\delta = 9/8$, $P(s, 9/8) = -s(4s-1)^2/8$ (where P is the sign-determining numerator of $f_{2b}(s,\delta) - 9/16$ with $s = 2 - 1/\eta$), so $f_{2b}(s, 9/8) = 9/16$ at the two points $s = 0$ and $s = 1/4$. The first corresponds to FPS's $\eta = 1/2$; the second is the witness $\eta = 4/7$ used in Lemma 5. At every other δ, the second root $s = 1/4$ continues to give strictly more than 9/16, by the perfect-square factorisation $P(1/4,\delta) = 12(\delta-9/8)^2$ of Lemma 5.

6 Numerical illustration

The proof of Theorem 1 is analytic; the witness $\eta = 4/7$ gives a closed-form lower bound on $F(\delta)$ via Lemma 5. For orientation we record numerical values of f_1, $f_{2a}^{\max}$, and the true $f_{2b}^{\max}$ (which is in general strictly larger than $f_{2b}(4/7, \delta)$, since the witness is not the argmax). The values were computed with SymPy 1.14 and a brute-force 1D maximisation of f_{2b} at 4001 uniformly spaced η values per δ. Selected values:

δ	$f_1(\delta)$	$f_{2a}^{\max}(\delta)$	$f_{2b}(4/7, \delta)$	$f_{2b}^{\max}(\delta)$	$F(\delta)$
1.10	0.5500	0.5713	0.5628	0.5757	0.5757
$\delta_1 \approx 1.114907$	0.5575	0.5575	0.5625	0.5654	0.5654
1.120	0.5600	0.5535	0.5625	0.5634	0.5634
$-3 + \sqrt{17} \approx 1.123106$	0.5616	0.5513	0.5625	0.5627	0.5627
1.124	0.5620	0.5507	0.5625	0.5626	0.5626
$9/8 = 1.125$ **(FPS)**	**0.5625**	0.5500	**0.5625**	**0.5625**	**0.5625**
1.150	0.5750	0.5374	0.5625	0.5750	0.5750
1.200	0.6000	0.5339	0.5650	0.6000	0.6000

The fourth column, $f_{2b}(4/7, \delta)$, is the witness value used in the proof of Theorem 1 via Lemma 5. It always exceeds 9/16 except at $\delta = 9/8$ where it equals 9/16 exactly, by the perfect-square factorisation $f_{2b}(4/7, \delta) - 9/16 = 12(\delta - 9/8)^2/[7(4\delta - 1)]$. The fifth column, $f_{2b}^{\max}(\delta)$, is the actual maximum of f_{2b} over $\eta \in [1/2, \eta_b(\delta))$, computed by direct maximisation; the arg max slides smoothly from $\eta \approx 0.69$ at $\delta = 1.10$, through $\eta \approx 0.59$ at $\delta = -3 + \sqrt{17}$, down to $\eta = 1/2$ at $\delta = 9/8$. The witness $\eta = 4/7 \approx 0.5714$ lies in this Case 2b interval throughout, since $\eta_b(\delta) > 4/7$ for $\delta < (41 + 7\sqrt{37})/66 \approx 1.266$, covering all of $(1, 5/4)$.

Remark 8. The naive optimisation that takes $f_{2b}^{\max}(\delta) = \delta/2$ at every δ (which is true *only* at $\delta = 9/8$) leads to the equation $\delta/2 = f_{2a}^{\max}(\delta)$ after substituting (6). Rationalising by squaring out $\sqrt{\delta}$ and $\sqrt{\delta - 1}$ gives a degree-5 polynomial in δ that factorises over $\mathbb{Q}$ as

$$-\tfrac{1}{4}(\delta - 1)(3\delta - 4)(\delta^3 + 3\delta^2 - \delta - 4). \tag{10}$$

The cubic factor is irreducible (no rational roots among $\pm 1, \pm 2, \pm 4$) with discriminant $229 > 0$, hence has three distinct real roots; its unique root in $(1, 5/4)$ is

$$\delta_1 \;=\; -1 + \frac{4}{\sqrt{3}} \cos\left(\frac{1}{3} \arccos \frac{3\sqrt{3}}{16}\right) \;\approx\; 1.114907.$$

This δ_1 would be the optimal δ if Case 2b's maximum were $\delta/2$ throughout. Since this hypothesis fails on $(1, -3 + \sqrt{17})$ (by Corollary 3), and since $\delta_1 < -3 + \sqrt{17}$, the cubic root δ_1 is not the true optimum. The numerical $F(\delta_1) \approx 0.5654 > 9/16$ in the table above confirms this directly.

7 Implications

Theorem 1 shows that the choice $\delta = 9/8$ is sharp within the FPS three-case optimisation. Any tightening of the Lemma 2.3 constant $c_{\mathrm{FPS}} = 9/16$ must therefore change something else in [3]. We outline three concrete routes.

1. **Replace Vizing–Gupta.** The bound $\chi'(H) \le \Delta(H) + \mu(H)$ enters via [3, Lemma 2.2(ii)]. The Goldberg–Seymour theorem (Chen–Jing–Zang [2]) bounds $\chi'(H)$ by

$\max(\Delta(H), \lceil \Gamma(H) \rceil)$, where $\Gamma(H)$ is the fractional chromatic index. If one can show that the semi-random multigraph H of [3] satisfies $\Gamma(H) \le (c' + o(1))k$ with $c' < 9/16$, then $9/16$ improves to c'. The relevant moment / concentration calculation for $\Gamma(H)$ is structurally similar to the one in [3, §3] but the analogue of Claim 3.7 needs to be redone.

2. **Sharpen the multiplicity bound.** FPS's Proposition 3.4 only requires $\mu(H) = o(k)$. If one can show $\mu(H) \le (m + o(1))k$ with explicit $m < 1$, then the Vizing–Gupta bound $\chi'(H) \le \Delta(H) + \mu(H)$ gives $\chi'(H) \le (\delta/2 + m + o(1))k$ at $\delta = 9/8$; this is worse than the current $(9/16 + o(1))k$ unless one can also show that $\Delta(H) \le (\delta/2 + o(1))k$ at $\delta < 9/8$ with controlled μ.

3. **Modify the construction.** The semi-random selection of U from a high-degree set $U(d)$ plus a uniform-random completion is the structural input that produces the cases of Claim 3.7. A different construction – for example, an adaptive threshold d that depends on w, or a non-uniform random completion biased away from common neighbourhoods – would produce a different optimisation problem with different constants.

In all three routes, the constant $9/16$ is what one has to beat. The threshold $\delta = 9/8$ comes along for the ride: as soon as the underlying argument changes, the choice of δ may also change, and the algebraic transition at $\delta = -3 + \sqrt{17}$ identified here ceases to be relevant.

Acknowledgements

This note grew out of an erroneous earlier attempt to improve [3, Lemma 2.3] by tuning δ. The first version proposed the cubic root $\delta_1 \approx 1.114907$ as the optimum; the error (silently assuming Case 2b's η-monotonicity at every δ) was identified during a referee-style audit and is recorded in Remark 8 and in the retracted draft `tighter_fps_RETRACTED.tex`. The witness identity of Lemma 5 – which turned the gap into a two-line analytic proof – emerged from inspecting the factorisation $P(s, 9/8) = -s(4s-1)^2/8$ at $\delta = 9/8$ and noticing that the second root $s = 1/4$ persists for all δ. Symbolic verification was performed in SymPy 1.14.

References

[1] M. O. Albertson. Posted at `http://www.openproblemgarden.org/op/crossing_numbers_and_coloring`, 2007/2009.

[2] G. Chen, G. Jing, and W. Zang. Proof of the Goldberg–Seymour conjecture on edge-colorings of multigraphs. Preprint, 2019. `arXiv:1901.10316`.

[3] J. Fox, J. Pach, and A. Suk. Immersions and Albertson's conjecture. In *41st International Symposium on Computational Geometry (SoCG 2025)*. LIPIcs vol. 332, Schloss Dagstuhl, 2025. `arXiv:2510.05893`.